Products of four ±1

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The numbers $a_1, a_2, \ldots, a_n$ are equal to 1 or -1 and satisfy

$$a_1a_2a_3a_4 + a_2a_3a_4a_5 + \cdots + a_na_1a_2a_3 = 0.$$

Show that n is a multiple of 4.